Proposal 3

Classification of defects in photovoltaic modules

Complexity: ★★ (Medium)

Motivation:

Thermal inspection of photovoltaic (PV) modules is a non-invasive technique used to assess the health and performance of solar panels. By capturing and analyzing the heat signatures emitted by PV modules, thermal inspections can detect anomalies such as hotspots, dust accumulation, or damaged cells. Hotspots, in particular, can indicate potential issues like cell degradation or electrical faults. This proactive approach to maintenance allows for the early identification of problems, reducing downtime and maximizing the overall efficiency and lifespan of solar installations. Thermal inspection plays a crucial role in ensuring the reliability and energy yield of PV systems, making it an essential tool in the field of solar energy management and maintenance.

Dataset:

- *Dataset Size:* 20000 images.
- *Annotations:* Each image belongs to one of 12 classes (cell, cell-multi, cracking, hot-spot, hot-spot-multi, shadowing, diode, diode-multi, vegetation, soiling, offline-module and no-anomaly).
- *Format:* image.jpg and JSON file with image path and respective anomaly label.

`https://github.com/RaptorMaps/InfraredSolarModules`
(Download Link)

Challenge:

1. Develop three AI-models to evaluate the status of the PV module using thermal signatures including:
   (a) Model 1: Binary classification (anomaly or no-anomaly);
   (b) Model 2: Classification with 11 anomaly classes (cell, cell-multi, cracking, hot-spot, hot-spot-multi, shadowing, diode, diode-multi, vegetation, soiling and offline-module);

(c) Model 3: Classification with 12 classes (cell, cell-multi, cracking, hot-spot, hot-spot-multi, shadowing, diode, diode-multi, vegetation, soiling, offline-module and no-anomaly).

2. Describe data augmentation techniques that were used.

3. Compare the results of your AI-model with at least 1 existing models (e.g., MobileNet, VGG, EfficientNet and ResNet). Use the following metrics to assess the quality of your implementation:

   - Accuracy (%) and F1-Score (%) of training and testing;
   - Confusion matrix;
   - Model complexity (# parameters).

4. Discuss the results, taking into consideration the following paper:
   `https://www.sciencedirect.com/science/article/pii/S0196890424006599`

References:

- `https://www.sciencedirect.com/science/article/abs/pii/S0263224123006991?via%3Dihub`
- `https://www.sciencedirect.com/science/article/pii/S0196890424006599`